



Solution Set

Question 1

$$A \times A \times A = \{(-1, -1, -1), (-1, -1, 1), (-1, 1, -1), (-1, 1, 1), (1, -1, -1), (1, -1, 1), (1, 1, -1), (1, 1, 1)\}$$

Question 2

Since

$$B \cap C = \emptyset$$

we have

$$A \times (B \cap C) = \emptyset$$

and

$$(A \times B) \cap (A \times C) = \emptyset$$

so part (i) is verified. Also

$$A \subset B, \quad C \subset D$$

so

$$A \times C \subset B \times D$$

Question 3

$$A \times B = \{(1, a), (1, c), (2, a), (2, c), (3, a), (3, c), (4, a), (4, c)\}$$

$$A \times A = \{(i, j) : i, j \in \{1, 2, 3, 4\}\}$$

$$\emptyset \times B = \emptyset$$

Question 4

The sketches are: unit square; integer lattice with natural x-coordinates and integer y-coordinates; horizontal segment at $y = 1$ from $x = 0$ to $x = 1$; unit cube; solid cylinder over the unit disk for height 0 to 1; and the four points

$$(\sqrt{2}, 2), (\sqrt{2}, -2), (-\sqrt{2}, 2), (-\sqrt{2}, -2)$$

Question 5

R is reflexive because all

$$(a, a), (b, b), (c, c), (d, d)$$

are present. It is symmetric because

$$(a, b)$$

and

$$(b, a)$$

are both present. It is transitive because the only nontrivial block is

$$\{a, b\}$$

and all pairs needed inside that block are present.

Question 6

The two equivalence relations are the identity partition and the one-block partition:

$$R_1 = \{(a, a), (b, b)\}$$

$$R_2 = \{(a, a), (a, b), (b, a), (b, b)\}$$

Question 7

Injective: if

$$f(n_1) = f(n_2)$$

then

$$2n_1 = 2n_2$$

so

$$n_1 = n_2$$

Not surjective: for example

$$(1, 0) \in \mathbb{Z} \times \mathbb{Z}$$

is not equal to

$$(2n, n + 3)$$

for any integer n.

Question 8

To prove injective, set

$$\frac{5x+1}{x-2} = \frac{5y+1}{y-2}$$

which gives

$$(5x+1)(y-2) = (5y+1)(x-2) \Rightarrow -11x = -11y \Rightarrow x = y$$

For onto, take any

$$u \neq 5$$

and solve

$$u = \frac{5x+1}{x-2} \Rightarrow x = \frac{2u+1}{u-5}$$

This x is not 2, so every codomain value is reached.

Question 9

$$R = \{(1, 6), (2, 7), (3, 8)\}$$

$$\text{Dom}(R) = \{1, 2, 3\}, \quad \text{Range}(R) = \{6, 7, 8\}$$

Question 10

$$R = \{(0, 5), (1, 6), (2, 7), (3, 8), (4, 9), (5, 10)\}$$

$$\text{Dom}(R) = \{0, 1, 2, 3, 4, 5\}, \quad \text{Range}(R) = \{5, 6, 7, 8, 9, 10\}$$

Question 11

$$R = \{(2, 8), (3, 27), (5, 125), (7, 343)\}$$

Question 12

For every integer a and every integer b,

$$a - b \in \mathbb{Z}$$

so

$$R = \mathbb{Z} \times \mathbb{Z}$$

Domain and range are both

$$\mathbb{Z}$$

Question 13

For

$$f(x) = -|x|$$

domain is

$$\mathbb{R}$$

and range is

$$(-\infty, 0]$$

For

$$f(x) = \sqrt{9 - x^2}$$

the domain is

$$[-3, 3]$$

and the range is

$$[0, 3]$$

Question 14

For f, the formulas overlap at $x = 3$ and both give 9, so each input has one output. For g, the formulas overlap at $x = 2$ but give

$$2^2 = 4, \quad 3(2) = 6$$

so g assigns two values to $x = 2$ and is not a function.

Question 15

$$\begin{aligned}(f + g)(x) &= 3x - 2 \\ (f - g)(x) &= 4 - x \\ \left(\frac{f}{g}\right)(x) &= \frac{x+1}{2x-3}, \quad x \neq \frac{3}{2}\end{aligned}$$

Question 16

Assuming

$$f(x) = \frac{1}{x^2 + 1}$$

and

$$\begin{aligned}g(x) &= 3x + 2 \\ (g \circ f)(x) &= \frac{3}{x^2 + 1} + 2 \\ (f \circ g)(x) &= \frac{1}{(3x + 2)^2 + 1}\end{aligned}$$

Question 17

Using

$$f(x) = ax + b$$

and the point

$$(0, -1)$$

gives

$$b = -1$$

Using

$$(1, 1)$$

gives

$$a - 1 = 1 \Rightarrow a = 2$$

Hence

$$f(x) = 2x - 1$$

Question 18

It is not a function from

$$\mathbb{Z} \rightarrow \mathbb{Z}$$

because the same first coordinate can have different second coordinates. For example

$$(0, 0) \in f \quad \text{from } (a, b) = (0, 0)$$

and

$$(0, 1) \in f \quad \text{from } (a, b) = (0, 1)$$

Question 19

Injective: if

$$2^m = 2^n$$

then

$$m = n$$

Surjective onto B is immediate because each element of B has the form

$$2^n$$

Thus

$$f^{-1}(2^n) = n, \quad f^{-1}(y) = \log_2 y \text{ for } y \in B$$

Question 20

$$f(\{1, 2, 3\}) = \{3, 8\}$$

$$f(\{4, 5, 6, 7\}) = \{1, 2, 4, 6\}$$

$$f(\emptyset) = \emptyset$$

$$f^{-1}(\{0, 5, 9\}) = \emptyset$$

$$f^{-1}(\{0, 3, 5, 9\}) = \{1, 3\}$$